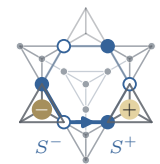
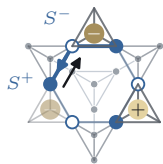


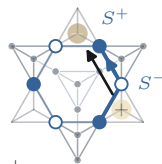
i.a) create


 $S_i^+ S_j^-$: pair splits

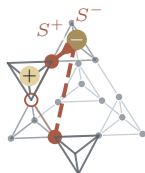
i.b) move


 $S_i^+ S_j^-$: spinon walks

i.c) annihilate

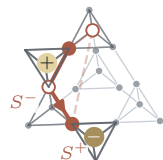

 $S_i^+ S_j^-$: + meets -
 $\mathbf{q}_\gamma = (0, 0, 0)$

i.d) create

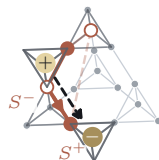
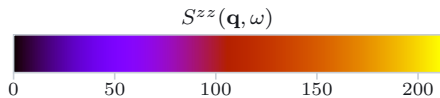


upper bond splits a pair lower bond: one period apart

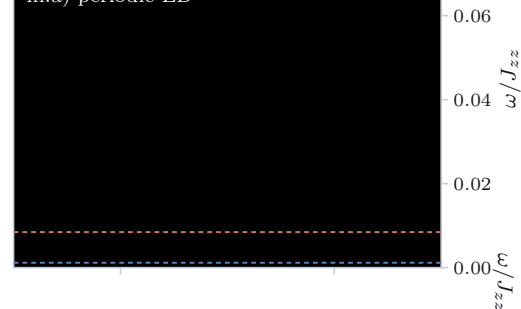
i.e) separate



i.f) annihilate


 the boundary closes it
 $\mathbf{q}_\gamma = (0, 0, -1)$


iii.a) periodic ED



iii.b) winding-free

